

KNOWLEDGE PACK

Learning Management Systems for Education





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LEARNING MANAGEMENT SYSTEMS FOR EDUCATION

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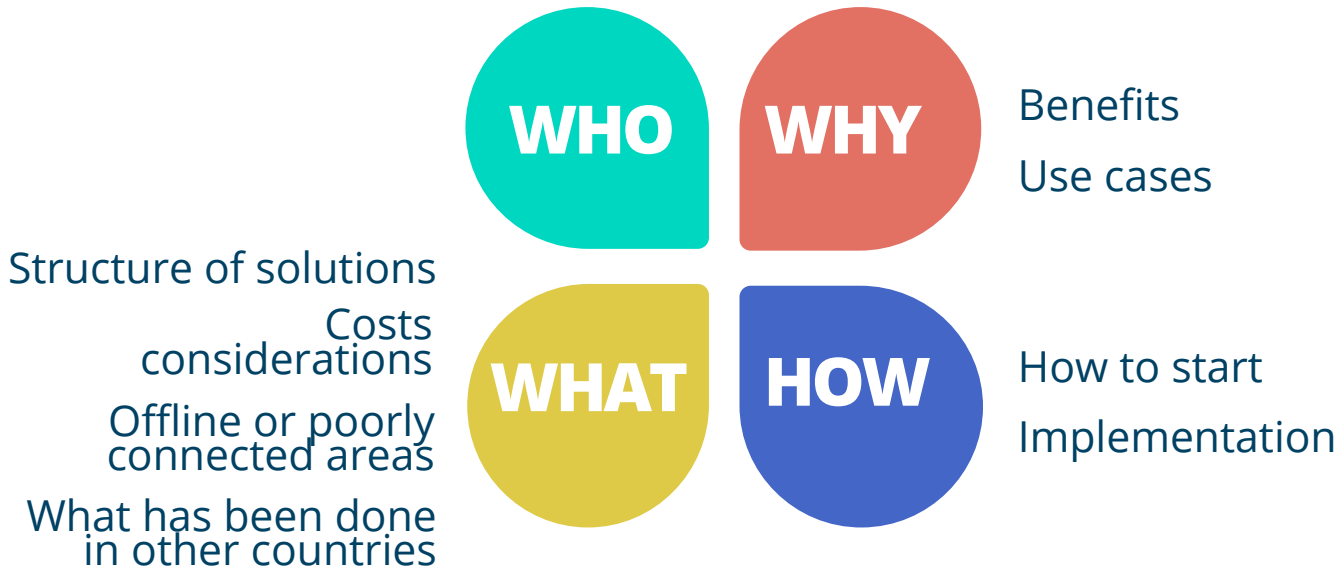
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Introduction

What is a KP?

Knowledge Packages (KPs) are short, pragmatic guides on individual topics within EdTech, meant to provide sufficient knowledge and understanding so that non-technical stakeholders can make key planning, design, and procurement decisions for education.

They can be used as a starting point for the planning of technology deployment to improve education, especially with education ministries.

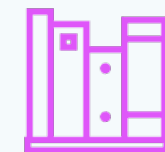
About this KP

The Learning Management Systems (LMS) Knowledge pack is meant to be a starting point in the decision-making process an entity could follow to decide what is the best course of action to choose and deploy a platform to support online training.

After reading the main content of this KP, some questions might pop* :



- What are the key considerations between open-source and proprietary LMS?



- What are some examples between open-source and proprietary LMS?

*To know more about these questions and their answers, go to the [FAQ available in the annex](#).

WHO are the main stakeholders ?



KPs are designed with a **human-centered vision**.

This knowledge pack is meant to provide sufficient knowledge and understanding so that non-technical Stakeholders can understand the fundamental elements involved in decisions about LMS for education.

Task Team Leaders (TTL's) & Bank Project Managers
(non-technical)



MOE Leadership
(non-technical)



MOE Program Managers
(semi-technical)



Donors, NGOs and Other Partners
(non-technical)



RESPONSIBILITY

- Assist MOE leadership in the application of KPs for in-country EdTech programs. Help design Bank-financed projects with practical information to include in project documents.
- Use KP to make key planning, design, and procurement decisions for in-country EdTech programs.
- Use KP to make key planning, design, and procurement decisions for in-country EdTech programs.
- Use KP to align with Bank EdTech programs and establish a common EdTech framework.



is this KP designed ?

[Benefits](#)[Use cases](#)

BENEFITS

Deploying a Learning Management System allows to:

- Organize and facilitate access to courses in a Digital format to students in any location
- Provide access to learning content that otherwise wouldn't be available (international sources)
- Support teachers' delivery of classes in synchronous and asynchronous manner
- Provide a platform for communication, collaboration between students and teachers
- Support the submission and grading of assignments and quizzes
- Record information about students' interaction with content, progress and engagement

- Monitor learning progress through summative and formative assessments
- Allow students to learn offline and in poorly connected environments.

A Learning Management System provides digital learning content to **support and engage students that cannot attend class or as a complement to face-to-face classes.**



is this KP designed ?

USE CASES

FULL OFFLINE /
LOW BANDWIDTH

- Packaged content
- Mobile client
- Synch data when connected

CONNECTED
REMOTE LEARNING

- Bandwidth appropriate content
- asynchronous learning
- Collaboration and project-based learning
- Grade books and gamification

SIMULATIONS AND
STEM LABS

- Rich practical simulations
- 3D support
- Multiuser collaboration
- Rich interaction and gamification

EXAMS /
PROCTORING

- In campus or remote examination
- High security user authentication

IN-CLASSROOM
LEARNING

- Rich interactive content
- Individual and group assignments
- Collaboration tools
- Grade books and gamification

Use case scenarios describe what we would like users to be able to do and their context



are the potential solutions?

STRUCTURE OF SOLUTIONS

KEY FEATURES OF AN LMS

A Learning Management System provides digital learning content and data to support and engage students that cannot attend class or as a complement to face-to-face classes. Here are the key features of an LMS:

Management of courses and users

- Organize & Display Courses
- Enrollment
- Monitor Progress

Management & organization of Content

- HTML5 – rich web content
- Video
- Documents
- Quizzes

Incentives to drive engagement

- Gamification
- Grades
- Leaderboards
- Badges

Collaboration & communication

- Forums
- Messaging
- Video conferencing (Webinars & Video lessons)

Learning Assessment

- Quizzes
- Assignments
- Exams
- Proctoring

Usage Reporting

- Data collection
- Visualization
- Analysis
- Integration with EMIS

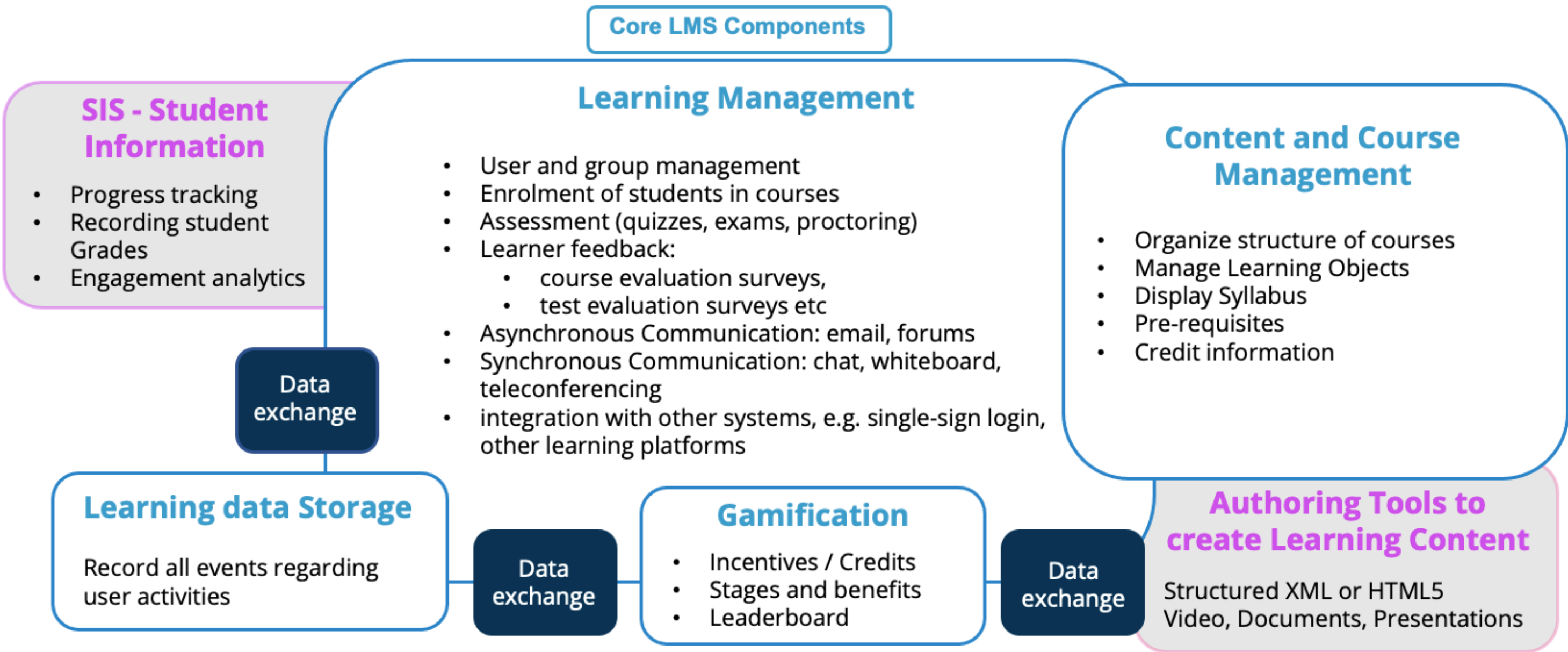


are the potential solutions?

STRUCTURE OF SOLUTIONS

ELEMENTS OF AN LMS

An LMS is at the core of an eLearning system and has to interact with several other systems that can be integrated or standalone.

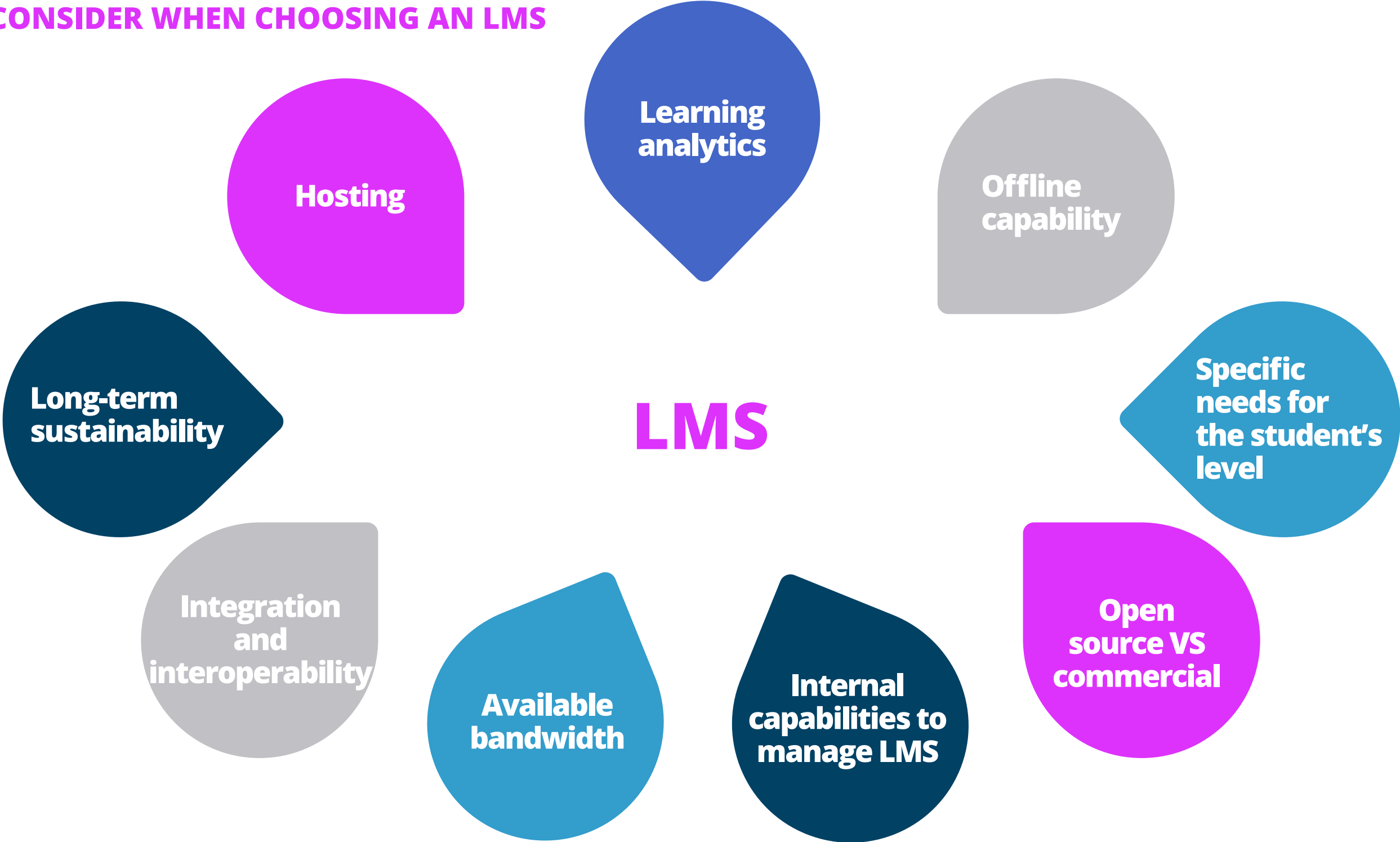




are the potential solutions?

STRUCTURE OF SOLUTIONS

FACTORS TO CONSIDER WHEN CHOOSING AN LMS





are the potential solutions?

STRUCTURE OF SOLUTIONS

DIFFERENCE BETWEEN OPEN-SOURCE AND PROPRIETARY

Open-source – the source code is developed by a community and is made available for download without any cost. It is possible to develop further the code and create custom functionalities. No support is provided.

Proprietary – the source code is developed and maintained by a company and its use requires a license which can be free or paid. Customization of code is done by the developer which usually also provides support. There are proprietary solutions developed on top of open-source code (ex. Open LMS, Redhat Linux)

Core LMS Software Code versus Cost of license	Open Source	Proprietary
License has no cost	Moodle Canvas Chamilo	Kolibri
Paid license	Open LMS (hosted Moodle) Canvas hosted	Blackboard Learndash (WP plug-in) Tribyte

With open-source code you are responsible for fixing bugs and adding features.
With proprietary code you would request support from the supplier.



are the potential solutions?

COSTS CONSIDERATIONS

BUDGETTING FOR LMS DEPLOYMENT

DEPLOYMENT OPTIONS

Commercial or open source?

Platform deployment

- Installation
- Customization
- Integrations

Platform maintenance

- Updating software
- Adding functionalities
- Fixing Bugs

Infrastructure maintenance

- Updating hardware
- Adding Hardware
- Increasing bandwidth

DYI or outsource?

Platform operation

- User management
- Content management

Content management

- Creating and maintaining courses
- Managing learning Objects

Type of hosting

- Inhouse datacenter
- IaaS
- SaaS

POTENTIAL COSTS TO CONSIDER

- Number of concurrent users as a % of total registered
- Licensing cost (per year for the amount of registered users or concurrent users)
- Hosting costs (IaaS or own platform)
- Setup/implementation and consulting fees
- Integration with other systems
- Data migration
- Platform updates and maintenance
- Technical Support
- Change management
- Skills training



are the potential solutions?

COSTS CONSIDERATIONS

TYPICAL PRICING MODELS

PER LEARNER, PER MONTH

The most common payment option is a per learner, per month subscription fee. This figure is either calculated based on the total number of users in the system or the number of active users enrolled in courses. Fees are generally \$5 or less per learner per month.

PER LEARNER, PER USE

LMS solutions can include more features, lessons, modules and tools than needed. A per learner, per use pricing model allows clients to opt out of certain parts of the software and pay only for what they use. This could mean being charged based on the modules in use, the number of active accounts in use or only the content that's being used by those accounts. Prices range from \$1 to \$10 per learner, per use.

PER COURSE

Some LMS vendors have a pay per course option for employees that need special certification for their industry or job function. Vendors might have content available in an internal library or partner with third-parties to deliver lessons through their LMS platform. Pricing is very variable.

LICENSING FEE

Companies can purchase a license to install LMS software on-premise. The licensing fee can be paid annually or upfront as a one-time fee. There are often additional implementation expenses for installation, data migration and training, and there may also be a charge per learner. Licensing fees generally range from \$500 to \$20,000, and setup can cost over \$25,000.

Four standard LMS pricing methods

[Structure of solutions](#)[Costs considerations](#)[Offline or poorly connected areas](#)[What has been done in other countries](#)

are the potential solutions?

LMS FOR OFFLINE OR POORLY CONNECTED ENVIRONMENTS

DEVICE BASED LMS CLIENT

When connectivity is not available or has poor quality, access to learning content and courses can be provided offline through either a local server or a LMS client installed in the user device which caches the content locally.

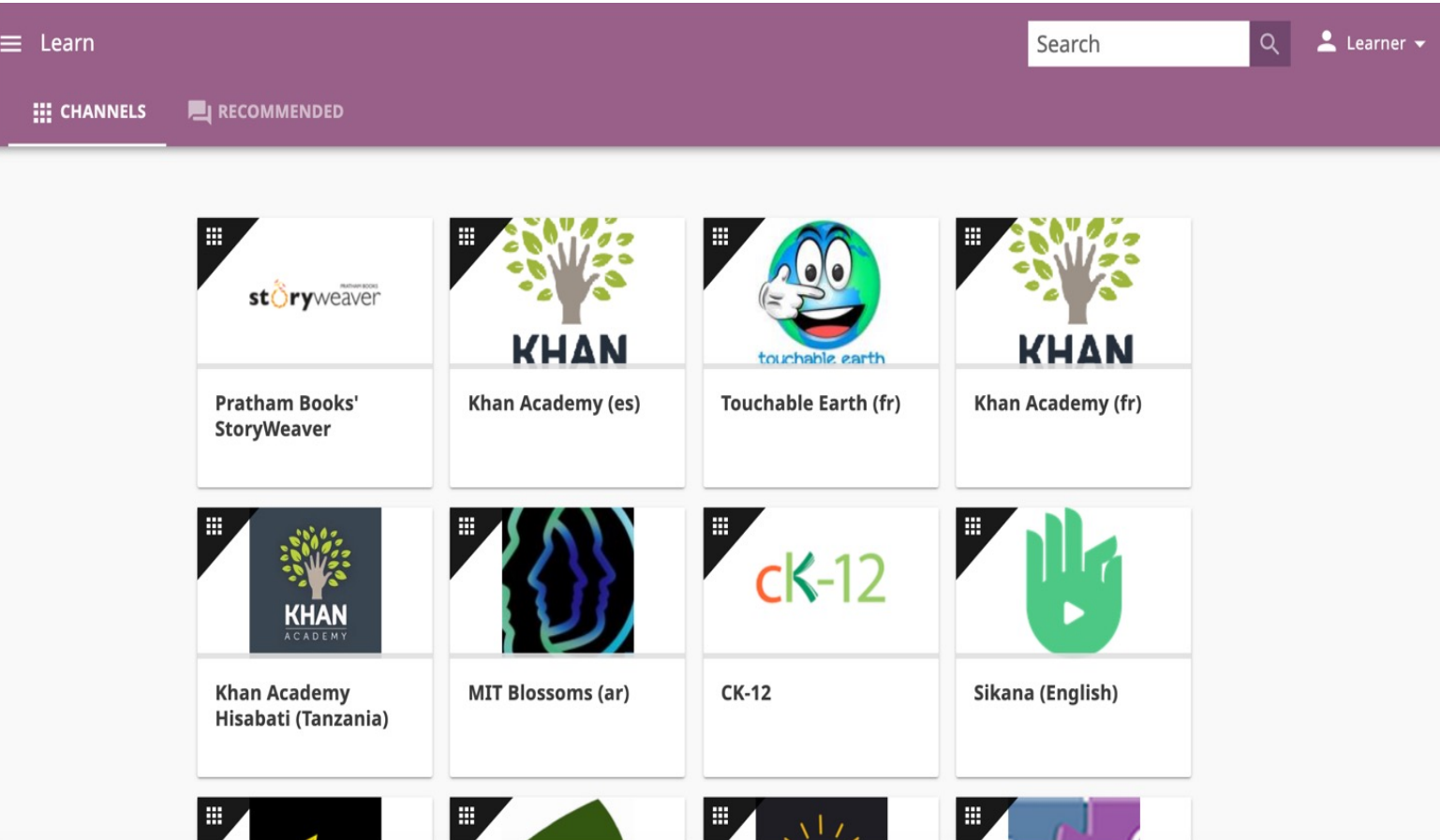
One of the most common options when connectivity is not available is to install the learning content directly into the learning device, especially when students need to use it at home.

This can be done by simply copying files into a structure of folder and by using an application that allows the content to be organized in a user interface.

Although this is a “better than nothing” option and something cheaper than printing and distributing books (especially if the devices already exist), it has proven to have severe limitations namely:

- 1. Difficult to update or add new content
- 2. No feedback provided on how students interact with the content
- 3. No interaction between the students, teachers and peers

Example of Kolibri





are the potential solutions?

LMS FOR OFFLINE OR POORLY CONNECTED ENVIRONMENTS

LOCAL LMS SERVER

When connectivity is not available or has poor quality, access to learning content and courses can be provided offline through either a local server or a LMS clients installed in the user device which caches the content locally.

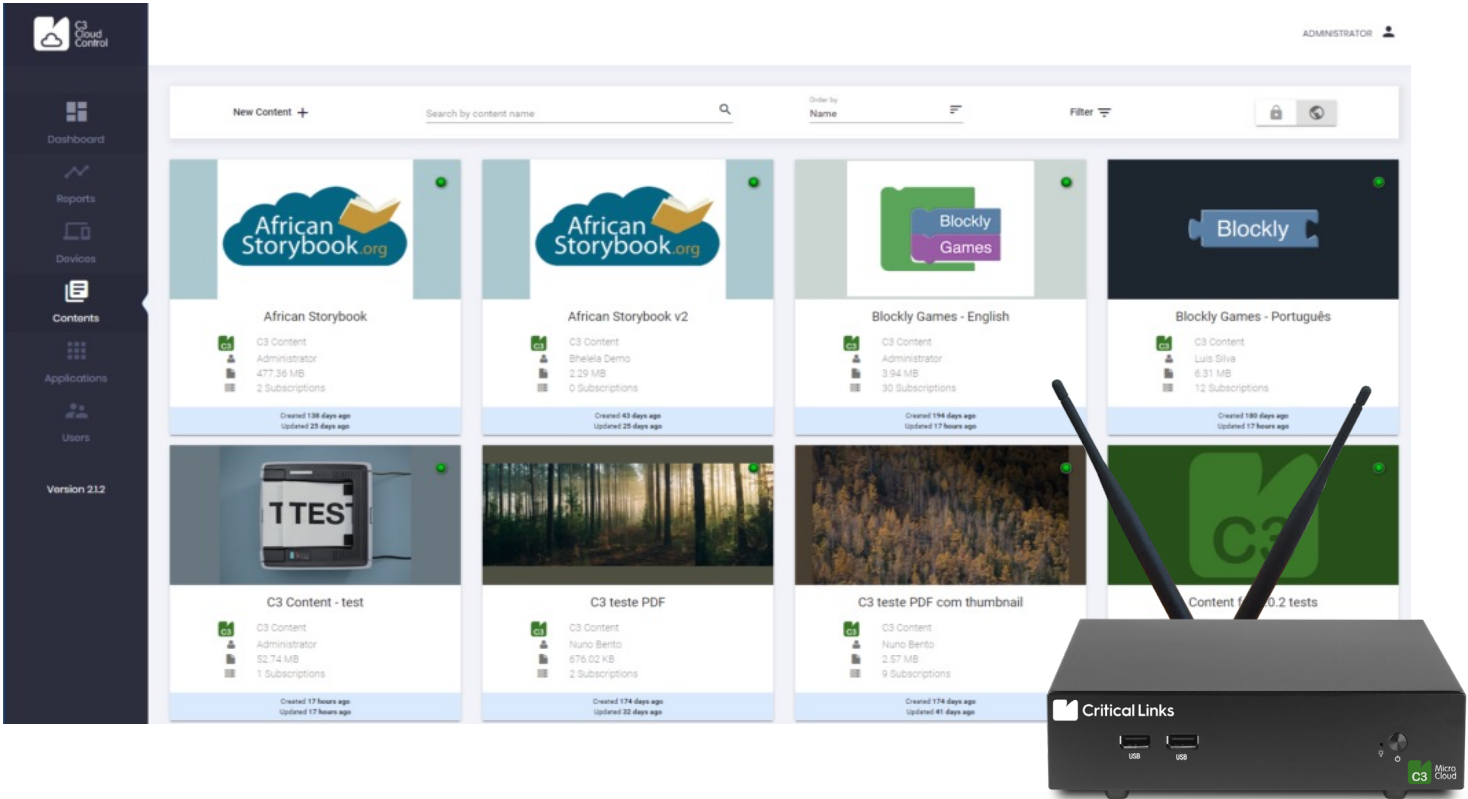
More and more projects are using “school Servers” to host learning content and shared applications.

These devices are installed in the school and provide access to learning content through a local instance of the LMS and a wifi connection provided by the device itself or a separate Wifi Access Point.

The server can be connected to internet through a mobile network or taken periodically to a connected location to get updates of content, applications but also upload student’s usage analytical data, providing important feedback to content managers.

This solution can also be a source of content for devices tat have a local LMS client allowing the update and replacement of content by simply having the student connecting to the school network.

Example of C3 Cloud





are the potential solutions?

WHAT HAS BEEN DONE IN OTHER COUNTRIES

EXAMPLES OF WORLD BANK FUNDED LMS IMPLEMENTATIONS

Vietnam Enhancing Teacher Education Program

Tunisia Digital Transformation for User-Centric Public Services

Turkey: Safe Schooling and Distance Education Project.



to implement next steps ?

[How to start](#)[Implementation](#)

HOW TO START

ESSENTIAL FEATURES TO TAKE INTO ACCOUNT FROM THE BEGINNING

Choose an LMS that is modular and able to seamlessly integrate with other solutions and share data in standardized format. Look for solutions compliant with one of the standards below.
There are 2 main entities defining data interoperability standards:

Advanced Distributed Learning Initiative

- SCORM (Sharable Content Object Reference Model)
- xAPI (Experience Application Programming Interface)

IMS Global Learning Consortium

- Caliper Analytics® Specification and Sensor API™ programming interface



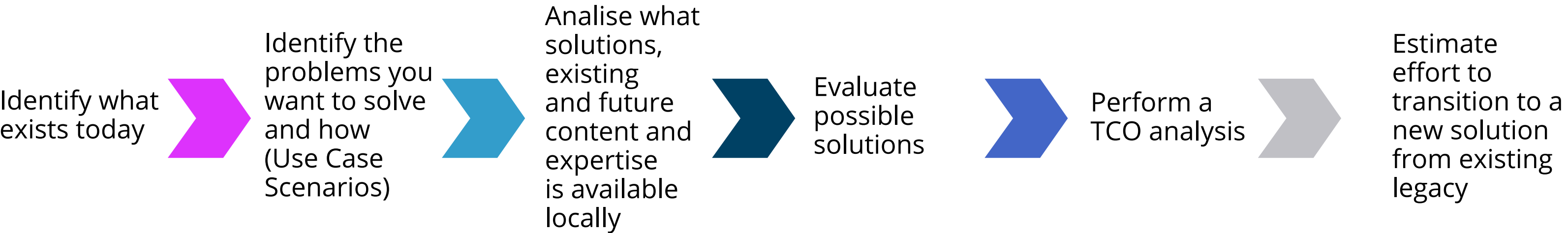
Around each of these specifications there is an ecosystem of applications and related specifications. Caliper is being developed by a closed consortium, and SCORM and xAPI in an open process.



to implement next steps ?

HOW TO START

WHERE TO START





to implement next steps ?

IMPLEMENTATION

KEY DECISION AND IMPLEMENTATION STAKEHOLDERS



POLITICAL

- MoE Top management
- MoE Finance department
- MoE Procurement
- HE institutions management
- EMIS administrators



TECHNICAL

- Infrastructure management team
- LMS management team
- Digital content development team
- EMIS administrators



PEDAGOGICAL

- Teachers
- Pedagogy team
- Curriculum development team
- Content development team



to implement next steps ?

IMPLEMENTATION

USE PROVEN APPROACHES – EXAMPLE OF PROFUTURO

Test before committing to a particular solution using MVP approach.



	January	February	March	April	May	June	July
Project plan (chronogram, team and dynamization plan)							
Conceptualization / Design of the MVP							
Solution development and server testing							
User Acceptance Test (design and execution)							
Online pilot development							
Evaluation reports							

Conclusion



Establish the team responsible for defining and implementing the LMS



Describe in detail the type of users and learning experiences you want to support

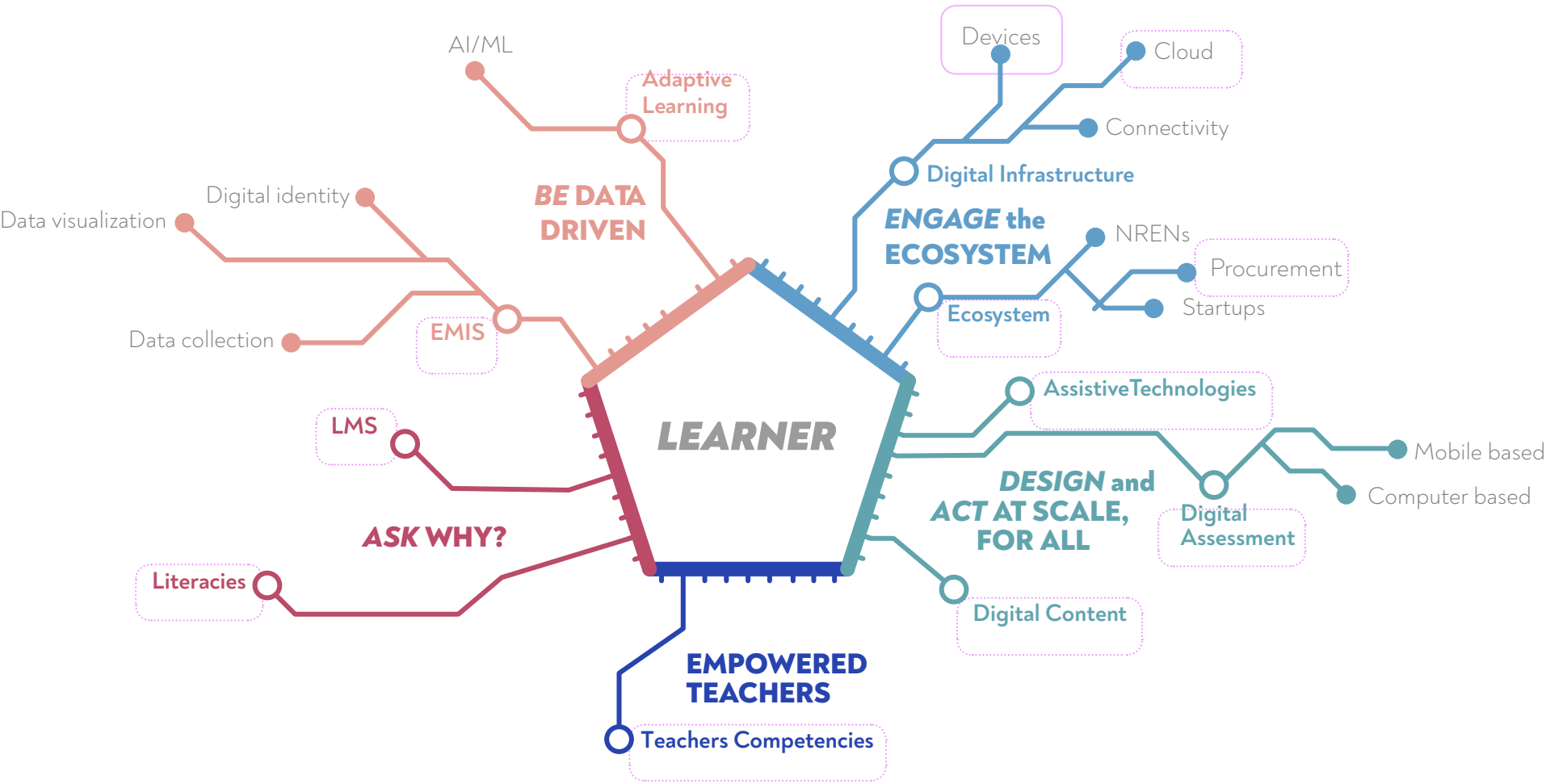


Choose your platform and implementation model.

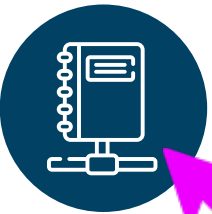


Establish your TCO budget and project plan

To go further
CLOUD OF KPs



OTHER EXISTING
RELATED KPs



EMIS 2.0



Digital content



Procurement



Ecosystems

RELATED SOURCES



For further resources refer to the [Annexes](#)

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Annexes

GLOSSARY

Adaptive learning is a segmentation of content and skills in modules so the student himself or in cooperation with the teacher and led by learning analytics, can choose what is the most productive to do next (progress or cover a learning gap, etc)

Assessment systems may be divided into two groups as:
Standalone systems without LMS, and Integrated systems with the LMS.

Asynchronous learning lets students access course materials at different times and locations. The main benefit is that users can learn at their own convenience. Because users aren't bound to a class schedule, organizations often set deadlines to motivate users to complete courses.

Caliper Analytics® is a standard from the IMS Global Consortium® which enables institutions to collect learning data from digital resources. Adopters of the standard can use Caliper to better understand and visualize learning activity and product usage data (1) .

Certification and compliance management – Some industries or specific jobs require employee certification or compliance training. This feature manages and tracks whether employees have completed all requirements and can generate reports for auditing.

Cloud hosting - The LMS is hosted on an infrastructure operated by a third party (e.g., Amazon, Microsoft or

Google) that operates and maintains the infrastructure to ensure that it's operating and online. The client maintains the software and data related to the LMS.

Commercial software requires the purchase of software licenses and generally does not afford users privileges to analyse source code, make modifications, or share the software.

Content development – Organizations can develop personalized course content, either using templates or from scratch. Course authoring tools let administrators design courses with multimedia, such as videos, PDFs, infographics, animations and PowerPoint presentations.



Annexes

GLOSSARY

Content library – Many LMS solutions include an out-of-the-box content library with premade learning materials. The library may cover a variety of topics, such as safety regulations, customer service and management techniques.

Gamification – Also known as game-based learning, gamification provides incentives for students to stay engaged by offering points, badges and other achievements. Students are motivated by competition, entertainment and the casual nature of games.

H5P makes it easy to create, share and reuse HTML5 rich and interactive web content and applications that is responsive and mobile friendly.

Learning analytics (LA) is a umbrella concept, defined by George Siemens (2012) and can be the tool for teachers to have an overview over learning successes and difficulties for a whole class of students, instead of having that overview for only a couple of them (if even that).

Learning Environments (LE) refer to the diverse physical or virtual locations, contexts, and cultures in which students learn.

Learning management helps users organize and automate education and training. It includes tools for managing user info, scheduling, alerts, messages and delivering content. Administrators can also set prerequisites for training and allow users to self-enroll into

specific courses.

Learning Management System (LMS) is a software application for the administration, documentation, tracking, reporting, automation and delivery of educational courses, training programs, or learning and development programs. LMS also provides the Course and Content Management System component and others.

Mobile learning gives students access to LMS materials through a mobile app or web browser. This feature allows remote and asynchronous learners to complete courses on the go.



Annexes

GLOSSARY

Open-source software is generally available to anybody to use freely (or at low cost), analyze source code, make modifications, and share under terms adopted by the software creator.

Performance tracking and reporting functions help determine if the overall learning program is effective. Organizations can analyse how users have performed in their courses and how much time they spend learning.

SaaS (or Software as a Service) -The supplier hosts the LMS, providing services such as customization prior to launch and trouble-shooting after the LMS is in operation. The client (MOE, university, school, etc.) licenses the software, typically paying an annual fee per user.

SCORM (Sharable Content Object Reference Model) was released in 2001 and is the most widely used standard in remote learning. SCORM-compatible content objects can be assembled into units of study that run in a browser-based LMS.

Self-hosting - The client owns and operates the infrastructure that host the LMS on premise or hosted externally.

Synchronous learning requires students to learn at specific times, either online or in a physical classroom, which can foster collaboration between employees. However, it can be difficult to coordinate and schedule synchronous learning.

Testing and assessments – Tests, assessments and quizzes see whether individual students have retained what they learned. They can be given after each lesson or at the end of the course. Analytical tools can identify proficiencies and learning gaps.

xAPI (Experience Application Programming Interface) was released in 2010 as a significant update to SCORM, one focused on improving information about a student’s achievement. xAPI supports recording information about a student’s progress through an online course and also student learning that takes place without a course and even without a computer.



Annexes

REFERENCES

EXTRA RESOURCES

- IMS Global - <https://www.imsglobal.org>
- Advanced Distributed Learning (ADL) - <https://adlnet.gov/>
- <https://elearningindustry.com/elearning-standards-scorm-aicc-xapi-cmi5-ims-cartridge>
- <https://ed.link/community/caliper-analytics/>
- Learning Analytics Community Exchange Comparing xAPI and Caliper Learning Analytics Review 7 ISSN: 2057By: Dai Griffiths and Tore Hoel - https://lace.apps.slate.uib.no/wp-content/uploads/2016/09/LACE-review07_xapi-caliper_v1-1.pdf
- How to include youth in low bandwidth areas - <https://www.s4ye.org/s4ye-publications>
- TCO - <https://www.webanywhere.com/2018/06/23/calculating-the-total-cost-of-ownership-for-totara-vs-a-proprietary-lms/> | <https://medium.com/@Software3E/open-source-vs-proprietary-lms-a-bird-view-c905c3218e5d>

Annexes

FAQ



- What are the key considerations between open-source and proprietary LMS?



- What are some examples between open-source and proprietary LMS?

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FAQ

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What are the key considerations between open-source and proprietary LMS?

Category	Open Source LMS	Proprietary LMS (Free or Paid)
Licensing	Free / Small cost for Software licensing.	Free or Annual licensing fee per user enrolment.
Features	Additional features added through plugins. These can be free, paid or require custom development.	Additional payment would be asked by vendors to initiate the development as per your requirements
Integrations	Some LMS offer integration through custom development	Integrations with leading solutions are usually available but may be offered at an extra cost.
Hosting	Usually hosted on infrastructure on prem or laaS	Hosted on own server or vendor server (SaaS)
Support	Support is usually provided by the community or paid service providers.	Tech-support is usually offered by the vendor or its official partners.
Scalability	Increasing users doesn't increase LMS licensing costs	Increasing users can increase licensing costs.
Security	Security fixes relies on open-source community and frequency of own maintenance and updates	Security checks and updates are provided by the vendor.
Keys Trade-offs	Free core LMS software but requires own hosting and technical skills to customize, enrich features and deploy. Sometimes support is limited to few knowledgeable people in the world. Some companies offer hosting and other services for a fee.	Usually offered as a SaaS, involves a set-up cost and a perpetual subscription license (per user basis) but includes hosting, updates, support and customization services. Can also be offered over an open-source platform.

Annexes

FAQ



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What are some examples between open-source and proprietary LMS?

Examples of Open-Source LMS's

- [Moodle™](#) : Moodle is an open source LMS free to download. Moodle is also offered as a subscription cloud solution and has several add ons such as Moodle App allowing courses to be done offline.
- [Chamilo](#) : Open LMS offers a highly curated, enhanced, SaaS-based versions of Moodle™ and full implementation and user support services.
- [Canvas](#) : Canvas is an open-source learning management solution created by Instructure. Instructure offers a reliable hosted and supported version for a set-up and licencing cost.

Is it important to note that Open Source doesn't always mean "free" software as some solutions based on open-source platforms are offered as a paid service.

[Google Classrooms](#) : Although it contains certain features of an LMS, Google Classroom is considered a Classroom Management system and not a Learning Management System.

Examples of Proprietary LMS's

- [Blackboard LMS](#) : Blackboard is a leading commercial proprietary solution. Fully compliant with IMS guidelines it provides many functionalities such as comprehensive analytics and engagement tools , assessments and certifications and collaboration tools, etc.
- [Learndash](#) : Learndash is a Wordpress LMS plug in that turns a WordPress site into a learning management system. It supports SCORM and xAPI and many other functionalities through third party plug-ins.
- [Open LMS](#) : Open LMS offers a highly curated, enhanced, SaaS-based versions of Moodle™ and full implementation and user support services.
- [Kolibri](#) : Kolibri is a free proprietary offline-first learning platform that runs on a variety of low-cost and legacy devices. Learning Equality offers customization and training for a fee.



INTRODUCTION

WHO

WHY

WHAT

HOW

CONCLUSION

ANNEXES

